Sustainability Management School Switzerland BUSINESS SCHOOL

CAPSTONE · AI FOR GOOD · WEEK 4

Scenario analysis: preparing a company for four possible futures instead of one forecast

The method behind Phase 1 of your report. Tonight: what a scenario is, why a company writes four of them, how to choose the two axes, and how to test Ørsted's and Eni's strategy in each world.

Dr. Jan Erik Meidell · Wednesday 4 November 2026 · online · 18:15–19:15, then the wind-farm economics · keep the shared canvas open on a second screen

TONIGHT, FIRST HOUR

Five parts and one exercise

1. **Why forecasts fail**: one number in the chat, Ørsted in 2023, the cone of uncertainty.
2. **What a scenario is**, and what it is not: forecast, sensitivity, contingency plan, scenario.
3. **Where the method comes from**: Shell in 1973, Schoemaker's ten steps.
4. **How to choose the two axes**: trends versus uncertainties, the impact × uncertainty grid, the independence check.
5. **Exercise**: each team's two axes on the screen, checked together.
6. **What a finished scenario contains**: a name, numbers, early signals, and a test of each company's strategy.

Break at 19:15. From 19:20: how an offshore wind farm makes, or loses, money (the second deck).

YOUR TURN

ONE NUMBER IN THE CHAT

How many gigawatts of offshore wind will the EU have in 2040?

Type one number in the Zoom chat. Do not look it up. Press Enter when I say "now", all at the same time.

The member states' own goal for 2040, agreed by sea basin (updated December 2024): **259–261 GW**. The point is not who was closest. The point is how far apart your numbers are: a room of informed people cannot agree on 2040, and neither can the industry.

What you just did is a **forecast**: one best guess of one future. Week 3 asked "what is the best estimate?". Tonight asks "what if the estimate is wrong?".

WHY A COMPANY BUILDS SCENARIOS

A plan built on one forecast fails when the forecast does

> **Scenario analysis** is a method in which a company writes several different but believable descriptions of the future (usually four), and then tests its strategy in each one: "would our plan still work if this happened?"

Nobody can predict Europe's electricity system in 2040: interest rates, climate policy, China, the grid are all open. A forecast gives one number; if the number is wrong, every decision built on it is wrong too.

A strategy that works in all four scenarios is **robust**. A strategy that works in only one is a **bet**. That is what the client pays you for in Phase 1: not a prediction of 2040, but a test of two companies' plans against futures they cannot control.

AN EXAMPLE · ØRSTED, 2023

Good forecasts, no scenarios: what it cost Ørsted

What happened

In autumn 2023 Ørsted cancelled two offshore projects off New Jersey, Ocean Wind 1 and 2, and wrote down DKK 28.4 billion (about €3.8 billion) in the first nine months of 2023, DKK 19.9 billion of it on Ocean Wind 1. A **write-down** (impairment) means the company declares that an asset is worth less than it paid. The dividend was suspended for 2023–2025.

Why

Four things moved at once: interest rates (up more than four percentage points in 2022–23), turbine and steel prices, the supply chain, and the timing of US permits and tax credits. Each was visible in 2021. Nobody had written them down together as one possible future and asked "what do we do if this is the world?"

Ørsted interim report, Q3 2023. A forecast asks what is most likely. Ørsted needed to ask what it must be ready for.

THE SAME MISTAKE, THREE COMPANIES

They had good data and one forecast; they lacked a second future

COMPANY	THE ONE FORECAST IT PLANNED ON	WHAT HAPPENED
Kodak	Digital cameras stay a niche (its own engineer built the first one in 1975)	Bankruptcy, 2012
Nokia	Smartphones stay a high-end niche above ordinary phones	Phone business sold to Microsoft, 2013–14
Ørsted	Offshore costs keep falling; the US market opens smoothly	US projects cancelled, 2023

They had

Good data, able analysts, careful sensitivity analyses, a strong central forecast.

They lacked

Several different futures written down · their strategy tested in each · signals to watch for · a plan for being wrong.

THE CONE OF UNCERTAINTY

The further ahead you look, the wider the range of possible futures

The cone of uncertainty: draw the possible values of a number, say Europe's wind capacity, year by year from today. For next year they sit close together; for 2040 they spread far apart. The shape is a cone opening from today. It shows where a single forecast still works and where it stops working.

2026–2028

1–3 years. The projects are already financed. A forecast works here.

2030

5 years. Auctions, interest rates and supply chains can still move the number a lot.

2040

15 years. Policy, technology and geopolitics can all change direction. One number here is a guess presented as an answer.

Your Week 3 learning curve is one line through the middle of this cone: the central estimate, not a certainty.

TWO DIFFERENT QUESTIONS

A forecast asks what is most likely; a scenario asks what we must be ready for

	FORECAST	SCENARIO ANALYSIS
The question	What is the most likely future?	Which futures must we be ready for?
The output	One number, with a margin of error	Three or four different, internally consistent descriptions of the future, each with numbers
Works for	Stable systems, 1–3 years ahead	Systems in transition, 10–30 years ahead
Fails when	The world changes regime and the forecast cannot see it	The stories are vague and change no decision

Your Week 3 projections are forecasts. They are inputs to the scenarios, not substitutes for them.

FOUR THINGS PEOPLE CALL "SCENARIOS"

Only the fourth one is a scenario

Forecast

One best estimate of one future. One line, one number: "wind supplies 38% of EU electricity in 2040".

Sensitivity analysis

One input changed at a time, everything else fixed: "what if the gas price is 20% higher?". It tests a model. It is not a scenario.

Contingency plan

A plan for one bad event, the "downside case". It still assumes one future.

Scenario

Several coherent futures, built on two independent uncertainties, each with a story and numbers, used to test a strategy.

Schoemaker (1995) draws this line. "Best case, base case, worst case" on one axis is a sensitivity analysis, not scenario analysis.

YOUR TURN

TYPE F, S, C OR SC IN THE CHAT

Which of the four is it?

1. "EU wind will supply 38% of electricity in 2040." **Forecast**
2. Our cost model run again with the build cost 15% higher, everything else unchanged. **Sensitivity analysis**
3. What Eni does if Plenitude's largest auction bid fails. **Contingency plan**
4. Four 2040 worlds from "EU policy ambition" × "openness of the supply chain", each with a story and numbers. **Scenario**
5. Best, base and worst case for the gas price. **Sensitivity analysis, with three names**

WHERE THE METHOD COMES FROM

Shell in 1973, and a distinction from 1921

Pierre Wack at Shell

From the late 1960s Wack's team at Royal Dutch Shell wrote scenarios instead of single forecasts. One of them described oil producers uniting and prices jumping. In October 1973 the oil embargo came and the crude price roughly quadrupled. Shell had already thought it through. Wack described the method in the *Harvard Business Review* in 1985.

Frank Knight: risk and uncertainty are different

Risk can be given a probability, because the past is a fair guide: the chance that a wind farm produces 5% less than expected in a given year. **Uncertainty** cannot: EU climate policy in 2040 has no track record to count. Scenarios are the tool for uncertainty.

Knight, *Risk, Uncertainty and Profit* (1921) · Wack, "Scenarios: Uncharted Waters Ahead", HBR (1985)

YOUR READING · SCHOEMAKER (1995)

Ten steps, and where each one happens in this course

STEP	WHAT YOU DO	WHEN
1–2	Define the scope · identify the stakeholders	Weeks 1–2, done
3–4	Identify the trends · identify the key uncertainties	**Tonight**
5	Build the scenario grid: two axes, four worlds	**Tonight**
6–7	Check consistency and plausibility · develop each scenario with numbers	Tonight → Week 5
8–9	Identify research needs · build quantitative models	Phase 2
10	Turn them into decision scenarios	Phase 3

Schoemaker, P.J.H., "Scenario Planning: A Tool for Strategic Thinking", *Sloan Management Review* 36(2), 1995, pp. 25–40.

STEPS 3 AND 4

Trends go into every scenario; uncertainties become the axes

A trend is a change you are fairly sure will continue. **An uncertainty** is a force that could go either way, and you cannot tell which.

Trends · fairly predictable

Electrification of transport and heating · turbines keep getting larger · old fossil plants close on known dates · climate targets already written into law · data-centre demand rising.

They go into every scenario, the same way.

Uncertainties · could go either way

Does the EU keep its climate line after 2030? Does the cost of wind keep falling, or stall? Do grid and permitting bottlenecks ease? Is the supply chain open, or used as a weapon? Do interest rates stay high?

Your two axes come from here.

The test: if you are fairly sure which way it goes, it is a trend, not an axis.

THE FILTER

Score every force on two things: how much it matters, and how unpredictable it is

↑ IMPACT ON THE QUESTION "IS WIND A SOUND INVESTMENT?"

High impact · low uncertainty

A trend. Put it in every world as a baseline. Example: electrification.

High impact · high uncertainty

A critical uncertainty. Your two axes come from this corner. Examples: EU policy after 2030; the pace of cost decline.

Low impact · low uncertainty

Ignore it. Example: the price of office paper.

Low impact · high uncertainty

Watch it, do not build on it. Example: blade-recycling technology.

UNCERTAINTY →

On the Week 4 page the sliders do this scoring for you: each force gets impact 1–5 and uncertainty 1–5, and the chart shows which ones land in the top-right corner.

STEP 5 · THE HARDEST JUDGEMENT

A good pair of axes passes three tests

1. **Truly uncertain.** Both ends of the axis must be believable in 2040. "Strong policy" and "weak policy" both have to be possible.
2. **High impact on the question.** The answer to "is wind a sound investment?" must be different at the two ends. If not, the axis is decoration.
3. **Independent of each other.** Knowing how one axis turns out must tell you nothing about the other. If axis B is partly caused by axis A, one corner of the grid cannot happen, and you have two or three worlds, not four.

The independence check (the technical word is "orthogonal", at right angles): write one sentence for each of the four corners. Look hardest at the two mixed corners, one axis high and the other low. If a corner can only be described with a far-fetched story, the two axes are not independent: keep the cause and replace the effect with another uncertainty.

BEFORE YOU COMMIT

Some forces move together, so they cannot be two axes

HOW STRONGLY ONE DRIVES THE OTHER	EU POLICY	INTEREST RATES	CARBON PRICE	COST DECLINE
EU policy	—	**weak**	**STRONG**	**moderate**
Interest rates	**weak**	—	**weak**	**STRONG**
Carbon price	**STRONG**	**weak**	—	**moderate**

EU policy and the carbon price cannot be two axes: the carbon price is set by policy, so the corner "strong policy, low carbon price" cannot exist. High interest rates raise the cost of capital and slow the fall in costs, so those two are linked as well. EU policy and interest rates are a good pair: central banks do not set rates to help wind.

Rule: make the second axis a **cause** of change, not an **effect** of the first axis.

YOUR TURN

EXERCISE · EACH TEAM'S AXES ON THE SCREEN

We check both teams' two axes together

1. I open your team's Week 4 block on the canvas. **One person** reads out: the two axes, each with its two ends, and the four corner names.
2. The **other team** writes in the chat one sentence for each of the two mixed corners (axis 1 high + axis 2 low, and the reverse): what Europe looks like in 2040 in that corner.
3. Together we decide: were both sentences believable? If yes, the pair **passes**. If one needed a far-fetched story, the pair **fails**: we name which axis is an effect of the other, and which uncertainty could replace it.
4. The scribe writes the result in the canvas box "The verdict on the axes": pass, or the axis swapped and what replaced it.

Team 1 first, then Team 2. If a team has no axes yet, we build them on the screen from the force list, in five minutes.

WHAT A PASS AND A FAIL LOOK LIKE

Two examples of axis pairs

Fails · the axes move together

"EU coordination" × "delivery capacity". A Europe that coordinates badly also delivers less, so the corner "poor coordination, high delivery" needs a long story to exist. The second axis is an effect of the first.

Passes · two independent causes

"EU policy ambition" × "pace of cost decline". A strong-policy world can still have slow cost decline if supply chains seize up; a weak-policy world can still get cheap turbines from global manufacturing. All four corners can happen.

Uncertainties that pair well with policy: the pace of cost decline · supply-chain geopolitics · interest rates · the speed of electrification · how fast the grid is built.

A WORKED GRID · FROM THE WEEK 4 PAGE

EU policy ambition × pace of cost decline gives four worlds

↑ FAST COST DECLINE

Market-driven greening

Weak policy, fast cost decline. Wind wins on price alone; new farms follow contracts with companies, not state auctions.

Renewable renaissance

Strong policy, fast cost decline. Both forces push the same way; the limit becomes ships, ports and people.

Stalled transition

Weak policy, slow cost decline. Cancellations and delays; gas stays the default. A pure wind company is in trouble.

Subsidised transition

Strong policy, slow cost decline. Wind is built because the state pays for it; profit depends on the contract.

WEAK POLICY ← → STRONG POLICY · ↓ SLOW COST DECLINE

STEP 7 · NUMBERS

Each world needs three to five numbers, called anchors

An anchor is a number for 2040 that pins a world down: the wind share of electricity, the cost per MWh, the carbon price. Without anchors a scenario is a mood; with them you can see whether the story holds together.

ANCHOR FOR 2040	RENAISSANCE	SUBSIDISED	MARKET-DRIVEN	STALLED
Wind's share of EU electricity	**55%**	**40%**	**35%**	**22%**
Offshore wind cost (€ per MWh)	**42**	**68**	**55**	**85**
EU carbon price (€ per tonne)	**180**	**160**	**60**	**45**

Illustrative numbers, to show the shape. Today wind is about 17–19% of EU electricity (Ember). Your own anchors need a source and a reason. A world called "fast cost decline" with a cost of €70 per MWh does not add up.

STEP 6 · THREE CHECKS, AND EARLY SIGNALS

Is each world a real scenario, and how would you know it is coming?

Consistent

Every detail fits the axes it sits on. "High carbon price with weak policy" fails: one causes the other.

Distinct

Four different worlds, not one story with four numbers. If all four are "growth", you have missed a possible break.

Plausible

Each world can unfold from where Europe is in 2026 without a miracle.

An early signal is something you could observe between 2026 and 2030 that tells you which world is arriving. For "Renaissance": offshore auctions clearing below €60 per MWh three rounds in a row; the EU's 2040 climate target written into law unchanged; a large offshore project finished on time and on budget. Two or three per world; in Phase 3 they become the client's early-warning dashboard (report §17).

FROM SCENARIOS TO STRATEGY

Put each company's strategy into each world and see whether it survives

Take a strategy and place it in each of the four worlds, like a model plane in a wind tunnel (the industry calls this "wind tunnelling"). In each world, does the strategy **thrive, survive, struggle or fail**? What breaks first?

Robust

Works in all four worlds. Less to gain, less to lose.

Flexible

Keeps options open: pay a little now for the right to commit later, when the signals are clearer.

A bet

Wins big in one world and fails in the others. Only with real conviction about that axis.

Ørsted · all on offshore wind

Sold its oil and gas. A bet: wins in "Renaissance", exposed in "Stalled" and whenever money is expensive. 2023 is what failing in one world looks like.

Eni · keeps oil and gas, grows renewables in Plenitude

Robust and flexible: rarely first, rarely last. Exposed in "Renaissance", where companies that went all in take the market.

INTO PHASE 1 · DUE SUNDAY 15 NOVEMBER, 23:59

What tonight gives your report, and what to avoid

MADE TONIGHT AND IN THE GROUP HOUR	WHERE IT GOES
The scored list of forces, and the trends	**Report §3** · the forces shaping the system
Your two axes, checked, and four named worlds with anchors	**Report §6** · the two uncertainties and the four worlds
Ørsted and Eni scored in each world (Week 4 page, exercise 3)	A first draft of **§15** (Phase 3)

Five mistakes to avoid: best / base / worst on one axis (a sensitivity analysis) · two axes that move together (two worlds, not four) · one world that is simply the EU target with three others to flatter it · no numbers · scenarios that never meet Ørsted and Eni.

Break until 19:20, then the second deck: how an offshore wind farm makes, or loses, money. Read Schoemaker if you have not: the ten steps (pp. 27–34) and the pitfalls (p. 39).